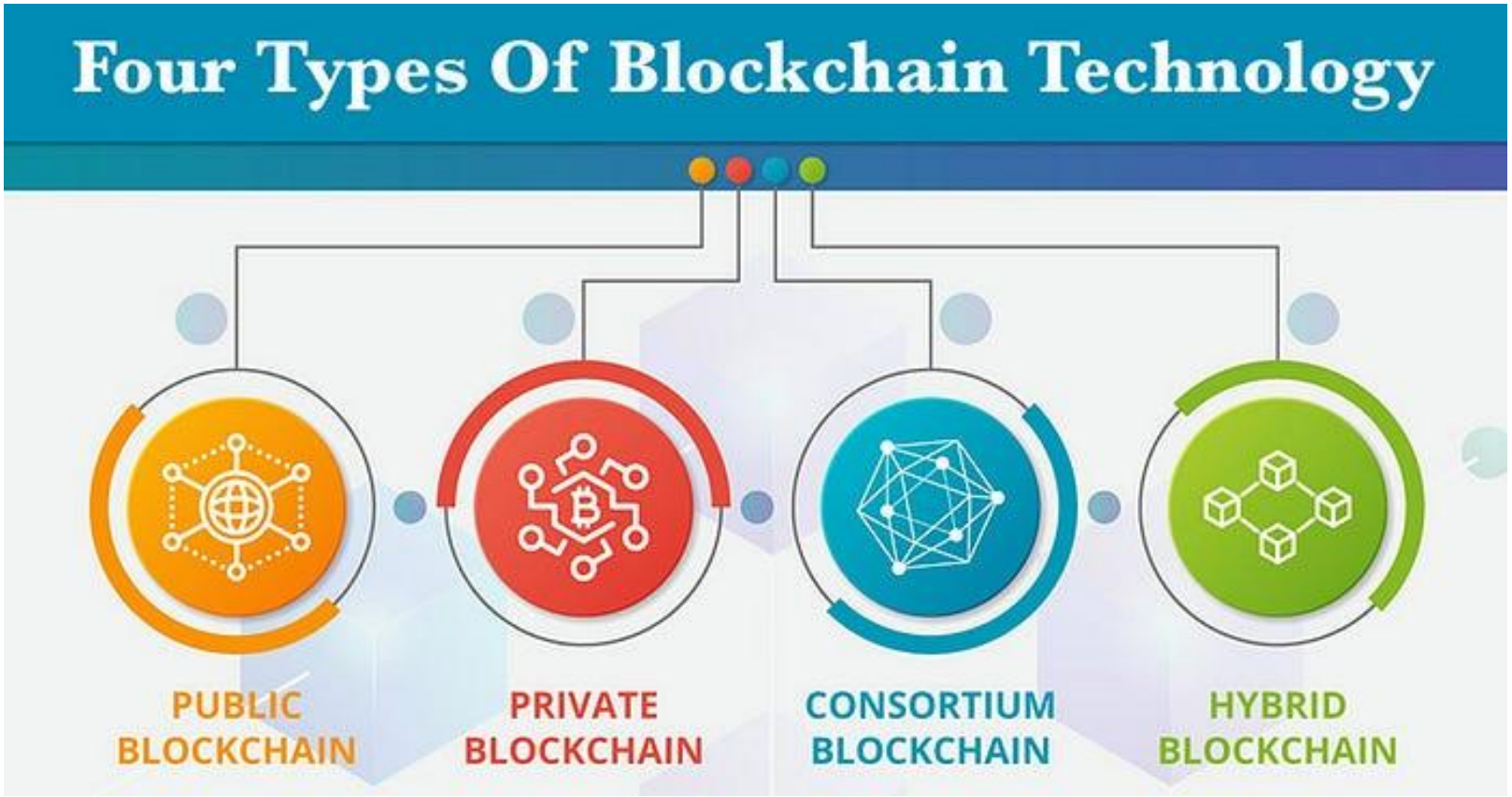
Types of Blockchain

Blockchain is a digital ledger technology that provides a secure and transparent way of storing and sharing data.

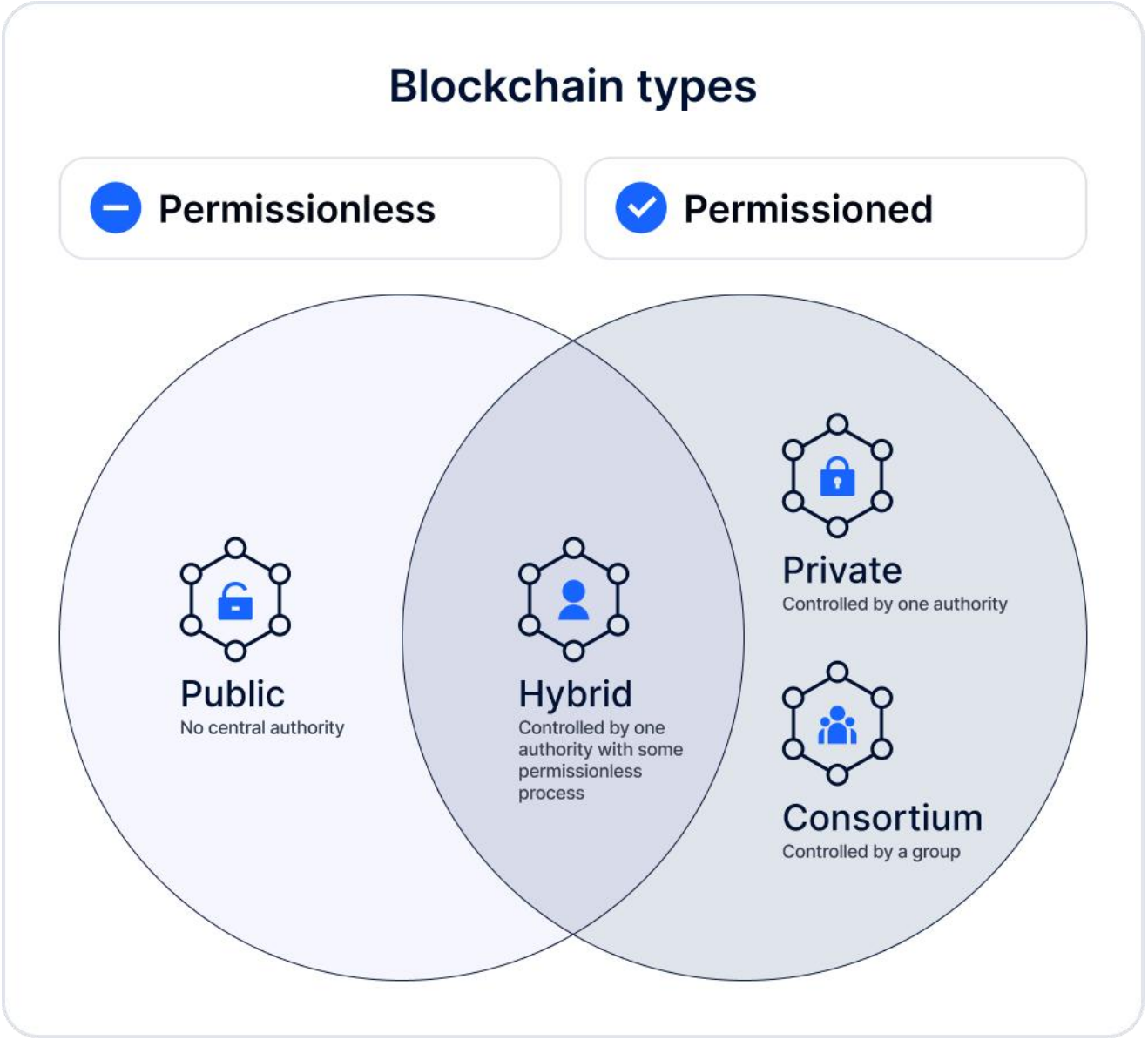
Types of Blockchain

- Public
- Private
- Consortium
- Hybrid



Types of Blockchain

Some of the blockchain types are categorized into permissioned and permissionless; they overlap as illustrated in the figure.



Public Blockchain

- A public blockchain is a type of blockchain technology that is open to anyone, and anyone with Internet facilities is eligible to participate.
- It Allows participants to validate blocks and send transactions without the need for permission from a central authority.
- They are characterized by their openness, transparency, and lack of central authority. Nodes can join and leave a network freely, and all nodes can verify new data added to the blockchain.
- They are permissionless, meaning anyone can access the blockchain without requiring permission, and the ledger is shared and transparent.
- Participants in a public blockchain can remain anonymous, as real names and identities are not necessary.

Public Blockchain

Characteristics of Public Blockchain

- Permissionless and open to all
- Cooperation among all nodes to verify data
- Use of incentive mechanism-based protocols
- Shared and transparent ledger
- Secure with 51% rule
- Anonymous and hidden user identity
- Absence of regulations or limitations on participant
- No way to track transactions

Private Blockchain

- Private blockchains are blockchain networks that are permissioned and accessible only to a specific group of people or organizations.
- Designed to restrict access to certain authorized users.
- Private blockchains are often used by organizations to build secure and private networks that can improve efficiency and reduce costs.

Characteristics of Private Blockchain

- Private blockchains operate in a closed network and have permissions managed by an organization.
- They are suitable for specific use cases where organizations want to exert control over access and network parameters.
- The advantages of private blockchains include faster transaction speeds, better scalability, and customization options.
- Private blockchains go against decentralization and distributed ledger principles.
- Use cases for private blockchains include supply chain management, asset ownership verification, and internal voting systems.

Public vs Private Blockchain

Basis of Comparison	Public Blockchain	Private Blockchain
Access	Permissionless	Permissioned
Network actors	Don't know each other	Know each other
Decentralized vs. centralized	Decentralized	More centralized
Order of magnitude	Lower	Higher
Native token	Yes	Not necessary
Speed	Slow	Fast
Transactions per second	Fewer	More
Security	More secure	Less secure
Energy consumption	More	Less
Consensus algorithms	Proof of work, proof of stake, etc.	Proof of elapsed time, raft, etc.
Attacks	Risk of collision or 51% attack	No minor collision, known validators
Effects	Disrupt current business models	Reduce transaction costs and data redundancies
Examples	Bitcoin, Ethereum, etc.	R3, EWF, B3i, Corda

Federated or Consortium Blockchain

- A federated blockchain is a type of blockchain network that operates under a federated consensus model.
- It involves a consortium of organizations or nodes that work together to validate transactions and maintain the blockchain.
- Federated blockchains are permissioned, meaning access and participation are restricted to authorized entities within the consortium.
- Federated blockchains are more centralized as the decision-making power lies with the participating entities

Characteristics of Federated Blockchain

- Permissioned and accessible only to authorized entities within a consortium
- Consensus achieved through a select group of nodes forming the federation
- Improved scalability and faster transaction speeds compared to public blockchains
- Enhanced privacy and security features
- More centralized decision-making compared to public blockchains
- Suitable for consortium-based collaborations and data sharing

Hybrid Blockchain

- A hybrid blockchain is a combination of both public and private blockchains, offering the benefits of both models.
- It allows for the interoperability of different blockchain networks and enables the exchange of data and assets between them.
- Hybrid blockchains provide flexibility in terms of transparency, control, and scalability.
- In a hybrid blockchain, certain parts of the network are public, allowing for open participation and transparency, while other parts are private, providing restricted access and enhanced privacy.

Characteristics of Hybrid Blockchain

- Combination of public and private blockchains
- Flexibility in terms of transparency and control
- Interoperability between different blockchain networks
- Customization of decentralization and privacy
- Suitable for applications with varying requirements

Blockchain Framework

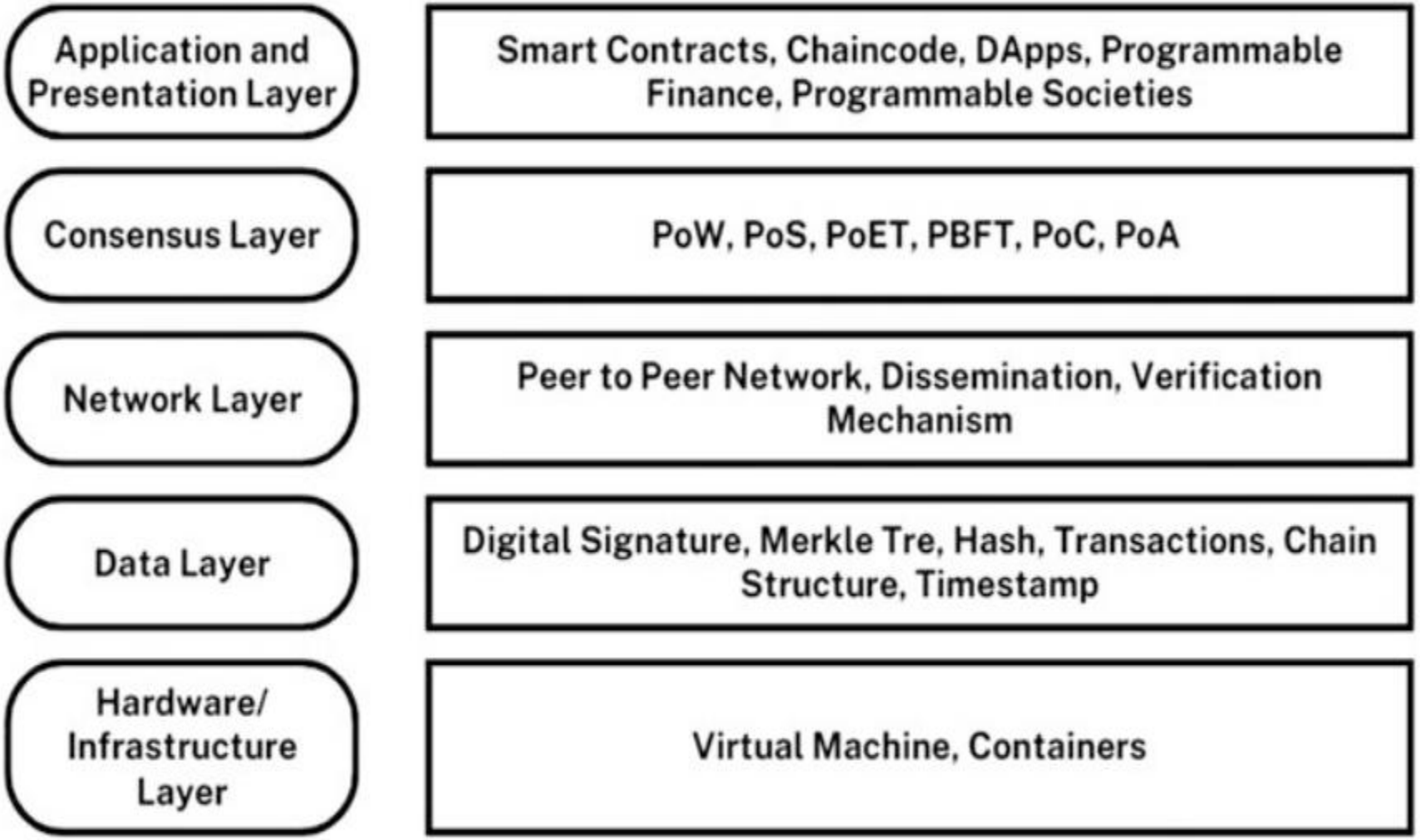
A blockchain framework is a set of protocols, rules, and standards that define the structure and operations of a blockchain network.

Blockchain Framework

Five layers in the Blockchain framework-

- Hardware/Infrastructure Layer
- Data Layer
- Network Layer
- Consensus Layer
- Application and Presentation Layer

Blockchain Framework



Hardware/Infrastructure Layer

- The bottom-most layer of the blockchain is the hardware or infrastructure layer.
- It involves servers hosted in data centers that store the content of the blockchain.
- These servers provide the necessary resources for the blockchain network to function effectively.

Data Layer

- By combining the use of linked lists, Merkle trees, and digital signatures, blockchain technology establishes a secure and transparent structure for storing and verifying data.
- The blockchain's data structure relies on a linked list, which consists of blocks with transactions and pointers to previous blocks.
- Pointers in the linked list refer to the location of other variables and maintain the sequential order of blocks.
- The blockchain employs a Merkle tree, a binary tree of hashes, to ensure security, integrity, and immutability.

Network Layer

- The network layer, also known as the P2P layer, is responsible for facilitating communication between nodes in a blockchain network.
- It enables transactions, block propagation, and discovery among the nodes.
- The network layer focuses on maintaining the validity of the blockchain network's current state by enabling effective communication, synchronization, and information propagation between nodes.

Consensus Layer

- The consensus layer validates blocks, ensures correct ordering, and achieves agreement among participants.
- In a distributed P2P network, the consensus layer establishes essential agreements and rules for maintaining integrity and security.
- Consensus ensures agreement on the validity and order of transactions, preventing manipulation and maintaining fairness.
- The consensus layer maintains power distribution and decentralization, preventing a single entity from controlling the blockchain.
- It enables collective decision-making and agreements among network participants.

Application and Presentation Layer

- The application and presentation layer is the user-facing part of the blockchain that provides a graphical user interface (GUI) and allows users to interact with the network.
- It includes execution layer and application layer protocols, such as smart contracts, scripts, and frameworks.
- Users can communicate with the blockchain network through various applications like wallets, social media apps, browsers, and NFT platforms.
- These applications interact with the blockchain network using application programming interfaces (APIs).
- The semantic layer within this layer is where transaction validations and executions take place, ensuring the integrity and accuracy of transactions.